An axial algebra whose automorphism group does not preserve its Frobenius form: an answer to a question of Shpectorov

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Abstract

In their survey of open questions on axial algebras, Gorshkov and Shpectorov record a question of Shpectorov: is there an axial algebra with a Frobenius form that is not preserved by the full automorphism group of the algebra? They ask for a non-trivial example, and especially for one whose space of Frobenius forms is 1-dimensional. We give a 4-dimensional example over the field $\mathbb{F}_{11}$. It is a simple algebra without identity, generated by four primitive axes for a C_2-graded fusion law on $\{1, 5, 7, 10\}$; every bilinear form that associates with its product is a multiple of one non-degenerate symmetric form β, and an automorphism of order 2 multiplies β by -1. The fusion law is not of Jordan or Monster type, and the field has positive characteristic; the question does not exclude either. Theorem 1 is formally verified in Lean 4. The example and its proofs were found and formalized by AI agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

We use the definitions of Gorshkov and Shpectorov [2, Section 2], which we recall briefly. Let A be a commutative, not necessarily associative algebra over a field $\mathbb{F}$. A *fusion law* is a set $\mathcal{F} \subseteq \mathbb{F}$ with $1 \in \mathcal{F}$ and a map $*$ from $\mathcal{F} \times \mathcal{F}$ to subsets of $\mathcal{F}$. For $a \in A$ and $\lambda \in \mathbb{F}$ let $A_\lambda(a)$ be the λ-eigenspace of the multiplication map $L_a \colon u \mapsto au$, and for $\Lambda \subseteq \mathbb{F}$ let $A_\Lambda(a) = \sum_{\lambda \in \Lambda} A_\lambda(a)$. A non-zero idempotent a is an *$\mathcal{F}$-axis* if $A = A_{\mathcal{F}}(a)$ and $A_\lambda(a)A_\mu(a) \subseteq A_{\lambda * \mu}(a)$ for all $\lambda, \mu \in \mathcal{F}$; it is *primitive* if $A_1(a) = \mathbb{F}a$. The algebra is a *primitive $\mathcal{F}$-axial algebra* if it is generated by primitive $\mathcal{F}$-axes. A *Frobenius form* is a non-zero bilinear form $(\cdot, \cdot)$ on A with $(uv, w) = (u, vw)$ for all $u, v, w \in A$.

Shpectorov asks [2, Problem 3.3]:

> Does there exist an axial algebra with a Frobenius form such that the full automorphism group of the algebra does not preserve the form?

The survey adds that the example should be non-trivial: an algebra $A = B \oplus B$ with a form that is non-zero only on the first summand does not count, because the automorphism exchanging the summands obviously moves the form. It is especially interested in algebras whose space of Frobenius forms is 1-dimensional; then each automorphism g multiplies the form by a scalar α_g, and for $g \mapsto \alpha_g$ to be non-trivial the algebra cannot have an identity element. In most known examples the space of Frobenius forms is 1-dimensional and the full automorphism group preserves the form [2]. Searches of the literature on 12 and 13 November 2026 (the second covering the arXiv listings announced on 13 November 2026) found no answer to the question,

and neither version of [2] marks it as answered; the coverage of the searches is described in the repository [1].

Let $\mathbb{F} = \mathbb{F}_{11}$ and let A be the 4-dimensional commutative algebra with basis a, b, c, d and product

$$a^2 = a, \quad b^2 = b, \quad c^2 = c, \quad d^2 = d,$$
$$ab = c + d, \quad ac = -b - d, \quad ad = b + c, \quad bc = a + d, \quad bd = -a - c, \quad cd = a + b.$$

Let $\mathcal{F} = \{1, 5, 7, 10\} \subseteq \mathbb{F}_{11}$ (note $10 = -1$) with the fusion law

$*$	1	5	7	10
1	1	5	7	10
5	5	$1, 5, 7$	$5, 7$	10
7	7	$5, 7$	$1, 5, 7$	10
10	10	10	10	$1, 5, 7$

(the entry in row λ and column μ lists $\lambda * \mu$). It is finite and symmetric, and it is C_2-graded with even part $\{1, 5, 7\}$ and odd part $\{10\}$ in the sense of [2, Section 2]. Let β be the bilinear form with $\beta(a, a) = \beta(c, c) = 1$, $\beta(b, b) = \beta(d, d) = -1$ and $\beta(x, y) = 0$ for distinct basis vectors x, y, and let g be the linear map that exchanges a with b and c with d.

Theorem 1. *(a) A is a primitive $\mathcal{F}$-axial algebra, generated by the primitive axes a, b, c, d.*

(b) β is a non-degenerate symmetric Frobenius form, and every bilinear form γ on A with $\gamma(uv, w) = \gamma(u, vw)$ for all u, v, w is a scalar multiple of β.

(c) g is an automorphism of A of order 2, and $\beta(gu, gv) = -\beta(u, v)$ for all $u, v \in A$.

(d) A is simple and has no identity element; in particular A is not the direct sum of two non-zero ideals.

So the full automorphism group of A does not preserve its Frobenius form, the space of Frobenius forms of A (together with 0) is 1-dimensional, and the character $g \mapsto \alpha_g$ of [2] takes the value -1 at g.

The mechanism is visible in the table: β gives the axes a and c length 1 and the axes b and d length -1, and g exchanges the two pairs. We have not determined the full automorphism group of A; one element that moves the form is enough.

Scope. The question asks only for an axial algebra, and the definitions of [2] allow any field and any fusion law; our example uses the field $\mathbb{F}_{11}$ and a fusion law chosen for it. The law is not of Jordan type $\{1, 0, \eta\}$ or Monster type $\{1, 0, \alpha, \beta\}$: it does not contain 0, and the multiplication map of each of the four axes is invertible. For algebras of Jordan type, [2] recalls a result of Hall, Segev and Shpectorov: such an algebra has a unique Frobenius form in which every primitive axis has length 1. Every automorphism preserves this form, so an example of Jordan type would need a space of Frobenius forms of dimension at least 2. We do not know whether there is an example of Monster type, or one over a field of characteristic 0. The axes all have 1-dimensional eigenspaces, so the fusion rules are a finite computation; the law above lists, for each pair of eigenvalues, exactly the eigenvalues that occur in the products.

Remark 1. The journal version of [2] adds, after the question, a remark of J. McInroy on the Miyamoto group of an algebra with a C_2-graded fusion law and the Frobenius form. In our example the Miyamoto involutions (the maps that are 1 on the even part $A_1(x) + A_5(x) + A_7(x)$ and -1 on the odd part $A_{10}(x)$ of an axis x) are $\tau_a = \tau_c$, which exchanges b and d, and $\tau_b = \tau_d$, which exchanges a and c; both preserve β, they generate a group of order 4, and g is not in it. This remark was checked by computer and is not formalized.

2 Proof of Theorem 1

All coefficients are in $\mathbb{F}_{11}$. Coordinates are with respect to the basis a, b, c, d.

(a) The axes. The four basis vectors are idempotents and form a basis, so they generate A. For each basis vector x, the following vectors are eigenvectors of L_x for the eigenvalues $1, 5, 7, 10$:

axis	1	5	7	10
a	$(1,0,0,0)$	$(0,1,7,1)$	$(0,1,5,1)$	$(0,-1,0,1)$
b	$(0,1,0,0)$	$(8,0,8,1)$	$(9,0,9,1)$	$(-1,0,1,0)$
c	$(0,0,1,0)$	$(7,1,0,1)$	$(5,1,0,1)$	$(0,-1,0,1)$
d	$(0,0,0,1)$	$(1,7,1,0)$	$(1,5,1,0)$	$(-1,0,1,0)$

This is checked by multiplying out. For each axis the four vectors have distinct eigenvalues, so they form a basis; hence L_x is diagonalizable with eigenvalues $1, 5, 7, 10$, each eigenspace is 1-dimensional, and $A_1(x) = \mathbb{F}x$. Write v_λ for the listed λ-eigenvector of a fixed axis x, so $v_1 = x$ and $v_1 v_\lambda = \lambda v_\lambda$. The other products, written in the basis (v_1, v_5, v_7, v_{10}), are

product	axes a, c, d	axis b
$v_5 v_5$	$(4,8,7,0)$	$(3,9,7,0)$
$v_5 v_7$	$(0,6,7,0)$	$(0,10,1,0)$
$v_5 v_{10}$	$(0,0,0,5)$	$(0,0,0,7)$
$v_7 v_7$	$(7,6,5,0)$	$(6,3,1,0)$
$v_7 v_{10}$	$(0,0,0,7)$	$(0,0,0,8)$
$v_{10} v_{10}$	$(2,4,8,0)$	$(2,6,7,0)$

(again by multiplying out). Since the eigenspaces are 1-dimensional, this gives $A_\lambda(x) A_\mu(x) \subseteq A_{\lambda * \mu}(x)$ for all $\lambda, \mu \in \mathcal{F}$, so each basis vector is a primitive $\mathcal{F}$-axis.

(b) The forms. Clearly β is symmetric and non-degenerate. Let $C(x, y, z) = \beta(xy, z)$ for basis vectors x, y, z. From the product, $C(a, a, a) = C(c, c, c) = 1$, $C(b, b, b) = C(d, d, d) = -1$, $C(x, y, z) = 0$ when exactly two of x, y, z are equal, and for three distinct basis vectors

$$C(a, b, c) = 1, \quad C(a, b, d) = -1, \quad C(a, c, d) = 1, \quad C(b, c, d) = -1,$$

the same for every order of the three arguments. So C is symmetric in its three arguments, and $\beta(xy, z) = C(x, y, z) = C(y, z, x) = \beta(x, yz)$; by linearity β is a Frobenius form.

Now let γ be any bilinear form with $\gamma(uv, w) = \gamma(u, vw)$; we do not assume that γ is symmetric. Fix a basis vector x and let y be another one. Then $\gamma(x, y) = \gamma(x^2, y) = \gamma(x, xy)$, and xy is plus or minus the sum of the two basis vectors other than x and y. For $x = a$ this gives, with $(p, q, r) = (\gamma(a, b), \gamma(a, c), \gamma(a, d))$,

$$p = q + r, \qquad q = -p - r, \qquad r = p + q,$$

which forces $2p = 2q = 2r = 0$, so $p = q = r = 0$. The same happens for $x = b, c, d$ (only the signs change), so $\gamma(x, y) = 0$ for all distinct basis vectors. Next, $\gamma(ab, c) = \gamma(a, bc)$ gives $\gamma(d, c) + \gamma(c, c) = \gamma(a, a) + \gamma(a, d)$, that is $\gamma(c, c) = \gamma(a, a)$; in the same way $\gamma(ab, d) = \gamma(a, bd)$ gives $\gamma(d, d) = -\gamma(a, a)$, and $\gamma(ac, b) = \gamma(a, cb)$ gives $\gamma(b, b) = -\gamma(a, a)$. Hence $\gamma = \gamma(a, a)\, \beta$.

(c) The automorphism. The map g is linear and $g^2 = 1$. It permutes the idempotents a, b, c, d, and on the six products of distinct basis vectors

$$g(ab) = d + c = ba, \quad g(ac) = -a - c = bd, \quad g(ad) = a + d = bc,$$
$$g(bc) = b + c = ad, \quad g(bd) = -b - d = ac, \quad g(cd) = b + a = dc,$$

so $g(uv) = g(u)g(v)$ for all u, v by bilinearity. Since g exchanges a basis vector of length 1 with one of length -1, $\beta(gu, gv) = -\beta(u, v)$.

(d) Simplicity and the identity. Fix a basis vector x. By (a), L_x is diagonalizable with eigenvalues $1, 5, 7, 10$, so $(L_x - 1)(L_x - 5)(L_x - 7)(L_x - 10) = 0$, which in $\mathbb{F}_{11}$ reads $L_x^4 - L_x^3 + L_x^2 + L_x = 2$. Hence L_x is invertible. The map $P_x = 3L_x^3 + 3L_x + 6$ is 1 on the eigenvalue 1 and 0 on 5, 7 and 10, so it is the projection onto $\mathbb{F}x$ along the other eigenspaces. In the table of (a), the eigenvectors of x for 5, 7, 10 have coordinate 0 at x; therefore $P_x(u) = u_x\, x$, where u_x is the coordinate of u at x. Now let I be a non-zero ideal and $u \in I$ non-zero, say $u_x \neq 0$. Since I is stable under L_x, it is stable under P_x, so $x \in I$, and then $A = L_x(A) \subseteq I$. Since also $A^2 \neq 0$, A is simple; in particular it is not a direct sum of two non-zero ideals. If z were an identity element, comparing the b-coordinate of $zb = b$, the d-coordinate of $zd = d$ and the c-coordinate of $za = a$ would give $z_b = 1$, $z_d = 1$ and $z_b + z_d = 0$, so $2 = 0$ in $\mathbb{F}_{11}$, a contradiction. □

3 Formal verification

Theorem 1 is formally verified in Lean 4 (version 4.34.1) with Mathlib [3] (version v4.34.1, commit `d13f23b7`). The files are in version 1.20.0 of the repository [1], directory `lean/Research/TaskG/`.

The statement. The statement file `Challenge.lean` contains only definitions and the statement, no proofs. It reuses the definitions of fusion law, axis, primitive axis, primitive axial algebra, ideal, Frobenius form and the product given by structure constants from the file `lean/Research/AxialMSZ/Challenge.lean` of an earlier paper in the same repository, which was reviewed and frozen then and is unchanged. The main statement, in ASCII notation, is

```
def WitnessConditions (F : FusionLaw K) (mu : V ->l V ->l V) (X : Set V)
    (beta : V ->l V ->l K) (g : V =l V) (c : K) : Prop :=
  F.IsFiniteSymmetric /\ X.Finite /\ X.Nonempty /\
  IsPrimitiveAxialAlgebra F mu X /\ DirectlyIndecomposable mu /\
  beta != 0 /\ (forall u v, beta u v = beta v u) /\ IsFrobeniusForm mu beta /\
  IsAlgebraAutomorphism mu g /\ c != 0 /\ c != 1 /\
  (forall u v, beta (g u) (g v) = c * beta u v) /\
  ((forall u, g (g u) = u) \/ (forall u, g (g (g u)) = u))
def ExampleOver (K : Type*) [Field K] : Prop :=
  exists n T F X beta g c, 2 <= n /\ n <= 8 /\
    WitnessConditions F (structProduct T) X beta g c    -- V = Fin n -> K
def ShpectorovAnswer : Prop :=
  ExampleOver Rat \/ ExampleOver (ZMod 5) \/ ExampleOver (ZMod 7) \/ ExampleOver (ZMod 11)
```

In this notation, bilinear products have the type `V ->l V ->l V`, invertible linear maps of V have the type `V =l V`, `IsAlgebraAutomorphism mu g` says $g(uv) = g(u)g(v)$, and `DirectlyIndecomposable mu` says that whenever $A = I \oplus J$ for ideals I, J, one of them is 0; this excludes every decomposition into two non-zero ideals, not only $B \oplus B$. `IsPrimitiveAxialAlgebra F mu X` says that the product is commutative, that every element of X is a primitive axis and that X generates the algebra. `IsFiniteSymmetric` asks that the fusion law be finite and symmetric, with values in $\mathcal{F}$; a primitive axis is a non-zero idempotent a with $A = A_{\mathcal{F}}(a)$, the fusion rules, and $A_1(a) = \mathbb{F}a$, as in Section 1. The bounds on n, the

four fields and the order of g only narrow the search and make the statement stronger. The stronger statement `StrongShpectorovAnswer` adds to `WitnessConditions` that every bilinear form γ (not assumed symmetric) with $\gamma(uv, w) = \gamma(u, vw)$ is a scalar multiple of β, and that the algebra has no identity element.

Review and freezing. An independent Codex agent, which wrote none of the files, reviewed the statement against the question and the definitions of [2] and approved it, and the file was frozen on 12 November 2026 at 16:27 UTC, before any proof file was written. The first compilation failed because Lean needs the facts that 5, 7 and 11 are prime to treat `ZMod 5`, `ZMod 7` and `ZMod 11` as fields; three lines providing them (each proved by `decide +kernel`) and a comment were added, nothing else changed (by then the proof files had been drafted; the change concerns only these instances), the same reviewer checked the difference, and the file was frozen again at 16:41 UTC with the SHA-256 hash

`05448636ff2c4fefdc731111eaf9659228123084ecc24525cfde978a37709bb8`

of `Challenge.lean`. The files were not changed when they were moved to the repository.

The proof. `Data.lean` defines A (as $\mathbb{F}_{11}^4$ with the structure constants above), β, g, the fusion law and the eigenvectors. The parts of Theorem 1 are the following theorems, in the namespace `TaskG.Proof`, proved along the lines of Section 2:

(a) `primitive_basis_axis`, `primitive_axial` and `fusion_finite_symmetric` in `Axes.lean`, and `fusion_grading` in `Main.lean`;

(b) `beta_symmetric`, `beta_frobenius` and `beta_form_space` in `Forms.lean`, and `beta_nondegenerate` in `Main.lean`;

(c) `g_mul`, `g_involutive` and `g_scales` in `Automorphism.lean`;

(d) `simple`, `directlyIndecomposable` and `hasNoIdentity` in `Simplicity.lean`.

`Main.lean` combines them into proofs of `ShpectorovAnswer` and `StrongShpectorovAnswer` with the field `ZMod 11` and $c = -1$. The frozen statements only say that some example exists; that it is the algebra of Theorem 1 is seen from the definitions in `Data.lean`, which `Check.lean` prints. Simplicity, the non-degeneracy of β and the C_2-grading are proved about this algebra in the proof files and are not part of the frozen statements. The finite computations use `decide +kernel` and `ring`. The audit file `Check.lean` restates both frozen statements and prints their axioms, and the proof files print the axioms of the theorems above: each list is exactly `propext`, `Classical.choice` and `Quot.sound`. No file uses `sorry`, `native_decide`, `bv_decide` or an `axiom` declaration.

Checks. On 12 November 2026 (17:25–17:32 UTC) ten files (the eight files of `TaskG`, the statement file of `AxialMSZ` that they use, and its tests) were compiled module by module, each by its own Lean process, in two new empty output directories on the author's Windows computer, with the pinned toolchain and Mathlib, which were not rebuilt; each module was then replayed with `leanchecker`. All 40 commands ended with exit code 0, with no error, no warning and no `sorry`, and the compiled `.olean` files of the two directories are byte for byte identical. From 12 to 13 November 2026 the whole Lean project of the repository, with these eight files, was rebuilt from a new empty directory on the author's Windows computer, module by module, each by its own Lean process, and not with `lake`: all 3019 modules compiled and 3018 passed `leanchecker`; the remaining one, the root of the project, has only imports and no constants. Some modules of another paper of the repository ran out of memory and were run again in a second pass; no Lean check rejected a module. The formalization has not been checked with comparator or with an independent kernel.

How this paper was produced

This work was carried out on 12 and 13 November 2026 (UTC) in a workflow directed by the author, using OpenAI Codex with the model GPT-6.1 Sol and Anthropic Claude Code with the model Claude Opus 5.5. Claude Code's scouting agents proposed the question, and Claude Code selected it and assigned it to Codex. Codex agents found the example by a computer search over a family of 4-dimensional algebras over $\mathbb{F}_5$, $\mathbb{F}_7$ and $\mathbb{F}_{11}$, wrote the statement file, the proofs and the Lean files, and ran the checks on the author's Windows computer; an independent Codex agent (same model), which had written none of the files, reviewed the statement, and another one gave the final review of the mathematics, the Lean files and the records of the checks, recomputing the example independently. Claude Code checked the example again by an independent computer calculation, searched the literature, rebuilt the whole Lean project of the repository and wrote this text; an independent Claude Code agent (model Claude Opus 5.5), which had written none of the files, reviewed this paper and the release before publication, and its corrections were applied. Throughout, "we" refers to this workflow; the reading and searching described in the paper were done by the agents. The author is not a professional mathematician. The author has read this paper and, using a side-by-side table, compared its Lean statement with Theorem 1. No human mathematician has reviewed this paper. The correctness of the result rests on the Lean proof and on the agreement of the Lean statement with the question of [2], which was checked by the reviewing agents; the written proof is given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

Thanks are due to Ilya Gorshkov and Sergey Shpectorov for the question and for collecting it with many others.

References

[1] H. Dong, *ai4math-results*, repository, https://github.com/Horace-Maxwell/ai4math-results, version 1.20.0 (2026); archived at Zenodo under the concept DOI https://doi.org/10.5281/zenodo.22970254.

[2] I. Gorshkov and S. Shpectorov, Axial algebras: questions and conjectures, *Commun. Math.* **35** (2027), no. 2, Paper no. 5, https://doi.org/10.46298/cm.19581; arXiv:2606.30048v1, 2026. We use the numbering of arXiv version 1: the question, attributed to S. Shpectorov, is Problem 3.3 there (in Section 3.1) and Problem 2.3 in the journal version, with the same wording.

[3] The mathlib Community, The Lean mathematical library, in *Proc. 9th ACM SIGPLAN Int. Conf. Certified Programs and Proofs* (CPP '20), ACM, 2020, https://doi.org/10.1145/3372885.3373824.